



23VSECCE401 - Programming Skills Development Laboratory

TOPIC :

PHANTOM FRAME

By

UCE2023403 – Ichcha Bhat

UCE2023404 – Riya Handu

UCE2023406 – Vaishnavi Ahire

UCE2023411 –Gauri Bankar

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Batch -A1

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1. Problem Statement

Develop an interactive Python-based application that integrates voice sentiment analysis, a real-time invisibility cloak effect, and a time-reversal video feature. The application should provide a user-friendly GUI, allowing users to interact seamlessly with these functionalities, enhancing user engagement and demonstrating the capabilities of Python in multimedia processing and natural language understanding.

2. Keywords

- Python
- Tkinter
- OpenCV
- Speech Recognition
- TextBlob
- Sentiment Analysis
- Invisibility Cloak
- Time Turner
- GUI Application
- Real-time Video Processing

3. Abstract

This project presents an innovative application that amalgamates natural language processing, computer vision, and graphical user interface design using Python.

The application offers three primary features:

1. Voice Sentiment Analysis:

Utilizing the speech_recognition library to capture user speech and TextBlob for analyzing the sentiment polarity, providing immediate feedback on the emotional tone of the spoken words.

2. Invisibility Cloak Effect:

Inspired by the fictional concept, this feature employs OpenCV to detect a specific color (e.g., red, blue, or green) in real-time video feed, replacing the detected area with the background, creating an illusion of invisibility.

3. Time Turner (Video Reversal):

Captures a short video segment and plays it in reverse, simulating a time-reversal effect, showcasing video manipulation capabilities.

The application is encapsulated within a modern GUI built with Tkinter, ensuring ease of use and accessibility. This project demonstrates the practical integration of various Python libraries to create an engaging multimedia experience.

4. Module-wise Description

a. Voice Sentiment Analysis Module

- **Functionality:** Captures audio input from the user, converts speech to text, and analyzes the sentiment.
- **Libraries Used:**
 - `speech_recognition`: For capturing and converting speech to text.
 - `TextBlob`: For performing sentiment analysis on the transcribed text.
- **Process:**
 1. Activate microphone and listen to user input.
 2. Transcribe audio to text.
 3. Analyze text sentiment (positive, negative, neutral).
 4. Display results in the GUI.

b. Invisibility Cloak Module

- **Functionality:** Creates an invisibility effect by replacing a specific color in the video feed with the background.
- **Libraries Used:**
 - `OpenCV`: For video capture and image processing.
 - `NumPy`: For numerical operations on image arrays.
- **Process:**
 1. Capture the background frame.
 2. Detect the specified cloak color in the video feed.
 3. Create a mask of the detected color.
 4. Replace the masked area with the background.

c. Time Turner Module

- **Functionality:** Records a short video segment and plays it in reverse.
- **Libraries Used:**
 - OpenCV: For video capture and playback.
- **Process:**
 1. Record video frames for a fixed duration.
 2. Store frames in a list.
 3. Reverse the list of frames.
 4. Play the reversed frames as a video.

d. Graphical User Interface (GUI) Module

- **Functionality:** Provides a user-friendly interface to access the application's features.
- **Libraries Used:**
 - Tkinter: For creating GUI elements.
 - ttk: For styled widgets.
- **Features:**
 - Buttons to activate each module.
 - Dropdown menu to select cloak color.
 - Display area for output messages and results.

5. Technology Selected

- Python: Chosen for its simplicity and extensive library support.
- Tkinter: Utilized for building the GUI, offering widgets and event handling.
- OpenCV: Employed for real-time video processing, including color detection and video manipulation.
- NumPy: Used for efficient numerical operations on image data.
- SpeechRecognition: Facilitates capturing and transcribing speech input.
- TextBlob: Provides tools for natural language processing, including sentiment analysis.

6. References

- GeeksforGeeks. "Invisible Cloak using OpenCV | Python Project." [Link](#)
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